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#include <ctype.h>
#include <stdio.h>
#include <string.h>

int n, m = 0, i = 0, j = 0;
char a[10][10], f[10];

void follow(char c);
void first(char c);

int main() {
    int z;
    char c, ch;

    printf("Enter the number of productions: ");
    scanf("%d", &n);

    printf("Enter the productions (example: E=E+T):\n");
    for (i = 0; i < n; i++) {
        scanf("%s", a[i]);
    }

    do {
        m = 0;
        printf("\nEnter the element whose FIRST & FOLLOW is to be found: ");
        scanf("%c", &c);

        first(c);

        printf("FIRST(%c) = { ", c);
        for (i = 0; i < m; i++) {
            printf("%c ", f[i]);
        }
        printf("}\n");
        strcpy(f, " ");
        m = 0;

        follow(c);

        printf("FOLLOW(%c) = { ", c);
        for (i = 0; i < m; i++) {
            printf("%c ", f[i]);
        }
        printf("}\n");
        printf("Continue? (1 = yes / 0 = no): ");
        scanf("%d", &z);
    } while (z == 1);
    return 0;
}

void first(char c) {
    int k;
    if (!isupper(c)) {
        f[m++] = c;
        return;
    }
    for (k = 0; k < n; k++) {
        if (a[k][0] == c) {
            if (a[k][2] == '$') {
                follow(a[k][0]);
            } else if (islower(a[k][2])) {
                f[m++] = a[k][2];
            } else {
                first(a[k][2]);
            }
        }
    }
}
```

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    }  
}  
  
void follow(char c) {  
    if (a[0][0] == c) {  
        f[m++] = '$';  
    }  
    for (i = 0; i < n; i++) {  
        for (j = 2; j < strlen(a[i]); j++) {  
            if (a[i][j] == c) {  
                if (a[i][j + 1] != '\0') {  
                    first(a[i][j + 1]);  
                }  
                if (a[i][j + 1] == '\0' && c != a[i][0]) {  
                    follow(a[i][0]);  
                }  
            }  
        }  
    }  
}
```